

ALTERNATIVE OPTION PRICING AND CVA.

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Abstract. The document IFRS 7 requires disclosure of information about the nature and extent of risks arising from trading those instruments. There are several significant drawbacks in derivative price modeling which relate to global regulations of the derivatives market. Here we present a unified approach which in stochastic market interprets option price as a random variable. Therefore spot price does not complete characteristic of the price in stochastic environment. Complete derivatives price includes the spot price as well as the value of market risk implied by the use of the spot price. This interpretation is similar to the notion of the random variable in Probability Theory in which an estimate of the random variable completely defined by its cumulative distribution function. If random variable is assigned to price and observations are interpreted as spot prices then correspondent cumulative distribution function is associated with buyer market risk. Therefore buyer market risk is the value of the chance that the spot price is higher than it is implied by market scenarios.

Options.

I. Let us briefly outline alternative to Black-Scholes benchmark [1,2] option pricing concept. European call option is a derivative contract in which buyer of the call option paying its price at initiation date receives the right to buy option's underlying asset for a known strike price K at maturity of the option contract T . In a simple discrete time setting with current date t and maturity T we denote $S(t) = x$ stock price at t and assume that the price of the stock $S(T, \omega)$ at T is a random variable taking values

$$S_1 < S_2 < \dots < S_m \leq K < S_{m+1} < \dots < S_N, \quad P\{S(T, \omega) = S_j\} = P(\omega_j) = p_j, \quad (1)$$

$j = 1, 2, \dots, N$. Call option price (premium) is a random function on t, x that depends on parameters T, K which is defined by equation

$$\frac{C(t, x; \omega)}{C(T, S(T, \omega))} \chi\{S(T, \omega) > K\} = \frac{x}{S(T, \omega)} \chi\{S(T, \omega) > K\} \quad (2)$$

and $C(t, x; \omega) = 0$ for a scenario ω if $S(T, \omega) \leq K$. Indeed, latter statement expresses the fact that there is no sense to buy call option for the scenarios for which market value of the underlying asset is bellow strike K . In this case buyer of the call option does not exercise the call option right to buy stock for K as he if he wishes can buy this stock on the market for the lower price than K . Solving (2) for the premium we arrive at the formula

$$C(t, x; \omega) = \begin{cases} 0, & \omega \in \omega_0 = \{ \omega : S(T, \omega) \leq K \} \\ \frac{x(S_j - K)}{S_j}, & \omega \in \omega_j = \{ \omega : S(T, \omega) = S_j > K \} \end{cases} \quad (3)$$

Equality (2) defines call option price with respect to underlying asset. This approach gives us no arbitrage pricing for each admissible market scenario ω . This formula is true for either deterministic or stochastic market, i.e. when probability of a particular scenario is close or tends to 1.

Given stochastic pricing (3) we interpret spot price of the option $c(t, x)$ as an estimate of the risk-reward or profit-loss ratios implied by the option. Therefore spot price of an instrument is its market risk representation. Indeed, risk-reward ratio can be represented as following

$$\frac{P\{C(t, x; \omega) < c(t, x)\}}{P\{C(t, x; \omega) \geq c(t, x)\}}, \quad \frac{\int_0^{c(t, x)} x dF_c(x)}{\int_{c(t, x)}^{+\infty} x dF_c(x)}$$

It is quite popular to use interpretation of the spot price $c(t, x)$ as ‘risk-neutral’ expected PV. We can admit this interpretation as a theoretical estimate of the market perception. It can be a good or not enough good theoretical estimate based on an assumption regarding underlying asset distribution.

Remark. The derivative pricing in the modern Finance Theory is interpreted as present value, PV of the cash flow, CF associated with financial contract. We consider CF as a formal definition of the real world transactions implied by the financial contract. Such price is usually referred to as no arbitrage price. The primary difference in interpretations of two definitions of the option price is what a security one considers as underlying to the contract. In (2) we consider the real stock as the underlying of the option contract. On the other hand using Black Scholes pricing interprets option underlying as the risk free bond.

Indeed, Black-Scholes concept can be briefly formulated as following: borrowing funds for buying option at risk free rate at t and return the borrowing sum plus its risk free interest at maturity T . Black and Scholes realize this idea by assigning to the option a heuristic stock

$$dS(t) = rS(t)dt + \sigma S(t)dw(t)$$

in which real expected return μS is replaced by the risk free expected return rS . The Black Scholes option price is called no arbitrage. Nevertheless, such defined no arbitrage option pricing does not eliminate market risk. It is easy to verify that applying no arbitrage option price buyer of the option is subject to market risk which implies a positive probability of getting loss at maturity. On the other hand a standard one step binomial scheme with two dates initiation date and maturity with arbitrary finite number of states of economy can not perform explicit solution of the option pricing problem.

Let c_0 denote spot price of the option at t . We interpret spot price as a settlement price between buyers and sellers at free market in contrast to theoretical price which comes from a particular theoretical strategy. The market risk of the European call option that is implied by the option premium c_0 is the chance that realized market scenario ω belongs to the set

$$\Omega_{\text{risk}}(c_0) = \{ \omega : C(t, x; \omega) < c_0 \} = \{ \omega : S(T, \omega) < \frac{Kx}{x - c_0} \} \quad (4)$$

where the value $v = \frac{Kx}{x - c_0} > K$ is a solution of the equation $\frac{v - K}{c_0} = \frac{v}{x}$. This equation follows from (2) in case when option price c_0 is given and we solving (2) for asset price $S(T, \omega) = v$. This value v corresponds to the option price c_0 . If the value of the underlying asset at maturity T will be below than v then this scenario is an element of the set $\Omega_{\text{risk}}(c_0; t, T, K)$ which by definition represents investor's market risk. Value of the premium c_0 depends on t .

II. The latest development of the derivatives theory pays attention to counterparty credit risk. Counterparty risk of the call option contract is the risk that seller of the European call option fails to deliver full value of the underlying assets at maturity of the contract. For example a default settlement may reduce the full contract delivery to its portion $q < 1$ if $S(T) - K \geq S_m = Q$. The problem is to present estimates of the values q and Q . On the other hand the default settlement can be also defined as

$$Q \chi \{ S(T, \omega) - K \geq Q \}$$

Suppose that M is defined by inequality $S_{M-1} - K < Q < S_M - K$. Then taking into account counterparty risk European call option payoff at maturity T can be defined by equality

$$C(T, S(T, \omega); \omega) = \begin{cases} 0, & \omega \in \omega_j, j \leq m \\ S_j - K, & \omega \in \omega_j, m < j \leq M \\ \text{a default settlement}, & \omega \in \omega_j, j > M \end{cases}$$

where a default settlement can be chosen by a portion $q = q(\omega) \in [0, 1)$ which is a recovery rate of the payoff $S(T) - K$ and by a constant Q . Barrier level Q is an important characteristic which represents the default recovery rate of the option seller. Bearing in mind formulas (1)-(3) we arrive at the market price of the option at t

$$C(t, x; \omega) = \sum_{j=m+1}^{M-1} \frac{x(S_j - K)}{S_j} \chi(\omega_j) + \sum_{j=M}^N \frac{xq(S_j - K)}{S_j} \chi(\omega_j) \quad (5')$$

$$C(t, x; \omega) = \sum_{j=m+1}^{M-1} \frac{x(S_j - K)}{S_j} \chi(\omega_j) + \sum_{j=M}^N \frac{xQ}{S_j} \chi(\omega_j) \quad (5'')$$

Given option price $C(t, x; \omega)$ it follows that implied values of the constants q and Q are equal to

$$q = \frac{EC(t, x; \omega) - \sum_{j=m}^M \frac{x(S_j - K)}{S_j} p_j}{\sum_{j=M+1}^N \frac{x(S_j - K)}{S_j} p_j}, \quad Q = \frac{EC(t, x; \omega) - \sum_{j=m}^M \frac{x(S_j - K)}{S_j} p_j}{\sum_{j=M+1}^N \frac{x}{S_j} p_j} \quad (6)$$

Hence in case when the barrier levels q or Q are known formulas (5) define stochastic option price. On the other hand in practice investors observe historical data for the stock and options and it is necessary to make an estimate of the barrier level. First formula in (5) holds two unknowns q and Q (M) while in the second formula in (5) there is only one unknown Q . Suppose that stock price is continuously distributed at T and

$$P(dS) = P\{S(T) \in [S, S + dS)\} = p(S) dS$$

where density $p(S)$ is a sufficiently smooth bounded function. From the first formula (5) it follows equalities

$$q = \frac{EC(t, x; \omega) - \int_K^{Q+K} \frac{x(S - K)}{S} P(dS)}{\int_{Q+K}^{+\infty} \frac{x(S - K)}{S} P(dS)}, \quad (7)$$

$$EC^2(t, x; \omega) = \int_K^{Q+K} \left[\frac{x(S - K)}{S} \right]^2 p(S) dS + q^2 \int_{Q+K}^{+\infty} \left[\frac{x(S - K)}{S} \right]^2 p(S) dS$$

This is a system of two equations that can be used for calculations q and Q . Then from the second formula (5) it follows that

$$Q = \frac{EC(t, x; \omega) - \int_K^{Q+K} \frac{x(S - K)}{S} P(dS)}{\int_{Q+K}^{+\infty} \frac{x}{S} P(dS)} \quad (8)$$

Equation (8) and the system (7) can be used to find for numerical solution. The default probability in the either case is reduced to $P(D) = P\{S(T) > Q + K\}$. Option default recovery rates q , Q are primary credit characteristics of the counterparty risk in the option contract.

III. In theory theoretical credit value adjustment, CVA to an option is defined as the difference between the risk-free option premium represented by (3) and the premium of the same option bearing in mind its chance of [counterparty](#)'s default represented by (5). Hence CVA represents possible losses of the bond buyer for each market scenario. Bearing in mind loss distribution we arrive at the fact that counterparty [credit risk](#) is a random process that changes its distribution during the lifetime of the option. Let

$C_0(t, x; \omega)$ denote stochastic price of the option given zero chance of counterparty risk. Then CVA of the call option is defined as following $C_0(t, x; \omega) - C_j(t, x; \omega)$, $j = q, Q$. Hence

$$\begin{aligned} & (1 - q) \sum_{j=M+1}^N \frac{x(S_j - K)}{S_j} \chi(\omega_j) \\ \text{CVA}(\omega) = & \left\{ \sum_{j=M+1}^N \frac{x(S_j - K - Q)}{S_j} \chi(\omega_j) \right\} \end{aligned} \quad (7)$$

where constants q and Q are defined in (6).

Remark. In [3] CVA at $t = 0$ is interpreted as [risk-neutral](#) expectation of the discounted loss

$$L^* = \chi(\tau \leq T) (1 - R) \frac{B_0}{B_\tau} E(\tau)$$

Hence

$$\text{CVA} = (1 - R) \int_0^T E^Q \left[\frac{B_0}{B_t} E(t) | \tau = t \right] d \text{PD}(0, t) \quad (8)$$

Here B_t is the future value of one unit of the base currency invested today $t = 0$, R is the recovery rate at the default time τ , $E^Q[\cdot]$ is the conditional probability of default taking with respect to risk neutral measure, $\text{PD}(0, t)$ is the date-0 risk neutral probability of default at t , and $E(t)$ is the exposure of a derivative contract (portfolio) at t . If derivative contract is represented in a single currency then $E_0 = 1$ and B_t can be thought as a standard discount factor. Exposure of the derivative contract is defined as

$$E(t) = \max \{ V(t), 0 \}$$

where $V(t)$ is the value of the contract at t . Note that from the buyer perspective the value of an option contract is always positive while for example the value of the swap can be either positive or negative. Therefore the option's exposure is equal to its value.

On the other hand if we interpret option price following (3), (4) CVA interpretation in the form (8) does not look appropriate. It is a common rule in the modern stochastic finance interprets the spot price as the risk neutral expected value of the correspondent cash flow. Such interpretation ignores the fact that expectation is a reduction of the stochastic cash flow that implies market risk of any theoretical pricing model including 'no arbitrage' even when the risk of default is equal to zero. In the formula (8) variables R , τ , $E(t)$ are interpreted as random depending on market scenario ω . Formula for CVA shows that the recovery rate R is assumed to be a nonrandom constant. This assumption significantly simplifies the problem setting. In order to take expectation in the formula for the discounted loss we need to integrate the right hand side with respect to joint distribution of the variables E and τ . Note that the credit risk theory deals with expected present value EPV of the cash flows. Assume for simplicity that default can occur only at maturity T , $\tau \geq T$. Then one equation represented by EPV contains two unknown variables

E and R. This is why the credit risk theory heuristically supposed that R is a known constant. One can calculate the probability of default assuming for example that R equal to 30% or other known number. Given stochastic formula for losses it is possible to present independent equations for higher order moments. These equations for higher moments admit calculations of the probability of default and nonrandom recovery rate simultaneously [4].

Next let us comment the risk neutral expectation in the formula (8). The underlying idea of using risk neutral valuation is coming from the Black Scholes formula. The BS call option price at t given that stock price at t is $S(t) = x$ is given by the formula

$$C(t, x) = B(t, T) E^Q \max \{ S(T; t, x) - K, 0 \} \quad (9')$$

One can ignore the use of risk neutral measure Q and state that regardless of the expected return μ of the real stock price given by equation

$$dS(t) = \mu S(t) dt + \sigma S(t) dw(t)$$

the BS price formula uses 'risk neutral' underlying $S_{RN}(t)$ having the risk free rate of expected return r given by equation

$$dS_{RN}(t) = r S_{RN}(t) dt + \sigma S_{RN}(t) dw(t)$$

defined on original probability space $\{\Omega, F, P\}$. The stock price process $S_{RN}(t)$ is heuristic, i.e. there is no stock which price is governed by the latter equation. Two random processes $S(t)$ and $S_{RN}(t)$ are defined on the same initial probability space $\{\Omega, F, P\}$ and formula (9') can be rewritten as

$$C(t, x) = B(t, T) E \max \{ S_{RN}(T; t, x) - K, 0 \} \quad (9'')$$

Let $t = 0$ denote the initial moment of time. Then the value option at a future moment $t > 0$ is specified by the value $S(t; 0, x)$ but not the value $S_{RN}(t; 0, x)$ which is only used for calculating BS option price at the point $(t, S(t; 0, x))$, $t > 0$. Hence the value of the option is equal to its exposure at a moment $t > 0$, i.e.

$$V(t) = C(t, y) |_{y=S(t; 0, x)} = C(t, S_{RN}(t; 0, x))$$

Therefore BS's call option price is a superposition of two random functions $C(t, y)$ and $S(t, \omega)$

$$\begin{aligned} C(t, S(t; 0, x)) &= B(t, T) E \{ E \max \{ S_{RN}(T; t, y) - K, 0 \} |_{y=S(t; 0, x)} = \\ &= B(t, T) [E C(T, S_{RN}(T; t, y))] |_{y=S(t; 0, x)} \end{aligned} \quad (10)$$

This equality shows that $C(t, y)$ depends on distributions of the S_{RN} on the interval $[t, T]$. The real stock process $S(t; 0, x)$ which is the underlying of the option should be considered as initial value at a future moment $t > 0$ and $(t, S(t))$ does not depend on whether it will apply for option valuation or not. The followers of the BS pricing should insist on using $S_{RN}(t)$ in formula (10) that represents the future value of the contract. The value of the call option contract $V(t) = C(t, S(t))$ is non negative and therefore exposure of the call option is equal to

$$E(t) = \max \{ V(t), 0 \} = V(t) = C(t, S(t))$$

Hence exposure $E(t)$ at a future moment $t > 0$ depends on the risk factor which is associated with stock price $S(t, \omega)$ defined with respect to measure P . Therefore there is a contradiction in using risk neutral expectation to calculate expected present CVA of the future exposure $E(t) = C(t, S(t))$. Bearing in mind that $\frac{B_0}{B_t}$ is a nonrandom known variable and equality

$$P\{E(t) \in [E + dE), \tau \in [t + dt)\} = P\{E(t) \in [E + dE) | \tau \in [t + dt)\} P\{\tau \in [t + dt)\}$$

we arrive at the formula

$$E[E(t) | \tau] = \int_0^T E[E(t) | \tau = t] dPD(0, t)$$

The latter equality coincides with (8).

IV. Now we will study effect of collateralization on option pricing. Collateralization of a financial contract changes the value of the transactions of the correspondent non-collateralized contract. These changes lead to the change in pricing and return characteristics of the contract. The reasoning in applying CVA and collateralization is outlined in [5]. In [6] one briefly presented primary approaches for pricing financial derivatives with collateralization.

Binomial case. Let us consider adjustments provided by collateralized option pricing in discrete scheme outlined in section 1. In classic binomial scheme which is based on BS option pricing this problem was studied in [7]. Let stock price be defined by formula (1) and time takes two values t, T , and $S(t) = x$. The value of the call option payoff at T is theoretically unbounded. Seller of the call option might default on delivery of the underlying stock or correspondent to stock cash if for some value Q the value $S(T) - K \geq Q$. Next we will ignore the difference in delivery of the underlying stock or the equivalent to the stock amount of cash. In (5'') we presented pricing formula which takes place regardless whether the pricing setting is discrete or continuous in time-space coordinate system. Therefore collateralization of the option represents pricing adjustment as well as a reduction of the risk exposure. Let $C_{cl}(t, x)$ denote call option price with collateralization at the date t . Assume that $x = S(t)$ and default barrier level Q is known. Payment $C_{cl}(t, x)$ represents initial value of the marked-to-market account MtM which is held by the option seller or a third party until maturity or default which one comes first. If default takes place at initiation date then premium $C_{cl}(t, x)$ would be returned to the option buyer. Dynamics of the MtM account can be described as following

$$MtM(t) = C_{cl}(t, x)$$

$$MtM(T) = [1 + i_{MtM}(t, T)(T - t)] C_{cl}(t, x)$$

Here $T - t$ is taking in an appropriate year format. Interest rate $i_{MtM}(t, T)$ which is applied to MtM account can differ from risk free interest rate. One can think about $i_{MtM}(t, T)$ as risk free bond rate, Libor, or Fed Fund rates. The popular families OIS or similar rates represent estimates of the risk free rate rather than the risk free rate itself. Indeed, one could not invest a sum in a swap rate and get \$1 in a particular future moment. If $S(T) - K < Q$ at T then there is no default and amount $MtM(T)$ goes to option seller while option buyer might exercise the right to buy stock $S(T)$ for K . If $S(T) - K \geq Q$ then seller of the option pays to option buyer amount of Q or equivalent portion of stock or other admissible by

contract assets. Formally call option collateralized contract from A and B perspectives can be defined by the cash flows

$$CF_A(t, T) = -C_{cl}(t, x)\chi(t) + \{[S(T) - K]\chi[S(T) - K < Q] + Q\chi[S(T) - K - MtM(T) \geq Q]\}\chi(T) \quad (10')$$

$$CF_B(t, T) = \{ \{ MtM(T) - [S(T) - K] \} \chi[S(T) - K < Q] - [Q - MtM(T)] \chi[S(T) - K - MtM(T) \geq Q] \} \chi(T) \quad (10'')$$

Here $\chi(t)$ is indicator which is equal to one at t and zero otherwise. The term $Q - MtM(T)$ on the right hand side (10'') shows that in case of default option seller uses MtM account to make payment of Q to option buyer. Collateralized option changes the cash flows which specify uncollateralized option contract. Using (10') it is easy verified that buyer of the option: pays at date t collateralized call option premium $C_{A,cl}(t, x)$ and receives call option payoff

$$C_{A,cl}(T, S(T)) = [S(T) - K]\chi\{0 < S(T) - K < Q\} - Q\chi\{S(T) - K - MtM(T) \geq Q\}$$

at date T . Hence the buyer's price of the option can be defined by the equation (2) that was used for uncollateralized price

$$\frac{C_{A,cl}(t, x; \omega)}{C_{A,cl}(T, S(T, \omega))} \chi\{S(T, \omega) > K\} = \frac{x}{S(T, \omega)} \chi\{S(T, \omega) > K\} \quad (2')$$

Seller of the option is structured differently. Seller does not received option premium at initiation date. The only transaction which takes place is defined by the right hand side of the equality (10''). It takes place at the moment T . The seller transaction is structured as the short position of the risky futures contract.

Assume for simplicity that interest rate $i_{MtM}(t, T)$ is equal to the risk free discount rate. Then solution of the equation $PV CF_A(t, T) = 0$ is less than the price of the uncollateralized option. If $i_{MtM}(t, T)$ is associated with the risk free discount rate then equalities (10'), (10'') lead to the equality of the buyer and seller option prices. Otherwise we arrive at the fact that buyer and seller option prices are not equal. If one wishes to take into account trading illiquidity factor the pricing in bid-ask format should be considered. Solving equation (2') we arrive at the formula

$$0, \quad \omega \in \omega_d = \{S(T, \omega) - K \leq 0\}$$

$$C_{A,cl}(t, x; \omega) = \left\{ \frac{x(S_j - K)}{S_j}, \omega \in \omega_j = \{0 < S(T, \omega) - K = S_j - K < Q + MtM(T)\} \right.$$

$$\left. \frac{xQ}{S_j}, \quad \omega \in \omega_0 = \{S(T, \omega) - K = S_j - K \geq Q + MtM(T)\} \right\}$$

which can be rewritten as

$$C_{cl}(t, x; \omega) = \sum_{j=M+1}^N \frac{x(S_j - K)}{S_j} \chi\{S(T, \omega) - K = S_j - K < Q + MtM(T)\} + \\ + \frac{xQ}{S_j} \chi\{S(T, \omega) - K = S_j - K \geq Q + MtM(T)\}$$

Collateralization could not fully eliminate counterparty risk. The chance of default is equal to

$$P\{S(T, \omega) - K \geq Q + MtM(T)\}$$

and average losses due to default can be represented by the formula

$$\sum_{S_j - K \geq Q + MtM(T)} \frac{x(S_j - K - Q)}{S_j} P\{S(T, \omega) = S_j\}$$

A spot price during the date t can be interpreted as a realization of the stochastic date- t option price. For example the risk-neutral Black Scholes estimate of the spot price $c_{cl}(t, x) = E^Q C_{cl}(t, x; \omega)$ implied the chance that call option is overpriced or underpriced by the market can be quantified by the probabilities

$$P\{c_{cl}(t, x) > C_{cl}(t, x; \omega)\}, P\{c_{cl}(t, x) < C_{cl}(t, x; \omega)\}$$

correspondingly. In the MtM pricing format option seller can avoid default for the scenarios when uncollateralized risky option (4) admits default. These market scenarios are specified by inequality

$$Q < S(T) - K < Q + MtM(T)$$

General case. Consider pricing adjustment provided by MtM collateralization. In continuous time default may occur at the end of each date when pricing adjustment takes place. Bearing in mind equations (2) and (2') we conclude that

$$C(t, x; \omega) = \sum_{l=1}^L \chi(\tau = t_l) \frac{xQ}{S(t_l)} + \chi(\tau > T) \frac{x[S(T) - K]}{S(T)} \quad (11)$$

Here indicator of the default $\chi(\tau = t_l)$ can be approximated by the expression

$$\chi(S(t_l) \geq Q) \prod_{i=1}^{l-1} \chi(S(t_i) < Q)$$

MtM adjustments take place at the end of each trading day. Let $t_{j+1} = t_j + h, j = 0, 1, \dots, N-1$ where constant h defines a MtM adjustment period which can be equal to one or any other appropriate number of days. Consider MtM transactions from the option seller perspective. Seller perspective is chosen as only seller makes adjustments to MtM account. We put

$$MtM(t_0) = C(t_0, x)$$

This equality reflects the fact that premium represents initial value of the MtM account. Next day just before new option price comes to the market the value of the MtM account is equal to

$$\text{MtM}(t_1 - 0) = [1 + i_{\text{MtM}}(t_0, t_1) \Delta t] \text{MtM}(t_0)$$

where $i_{\text{MtM}}(t_0, t_1)$ is an interest rate used by buyer and seller of the contract for MtM deposits. MtM account is set to be equal to market price of option at each moment prior to default or expiration moments which one comes first. Thus for a moment t_l , $t_l < \tau$ we set

$$\text{MtM}(t_l - 0) = [1 + i_{\text{MtM}}(t_{l-1}, t_l) \Delta t] \text{MtM}(t_{l-1}),$$

$$\text{MtM}(t_l) = C(t_l, S(t_l); \omega) = \text{MtM}(t_l - 0) + \delta_{\text{MtM}}(t_l),$$

where

$$\delta_{\text{MtM}}(t_l) = C(t_l, S(t_l); \omega) - C(t_{l-1}, S(t_{l-1}); \omega) [1 + i_{\text{MtM}}(t_{l-1}, t_l) \Delta t]$$

represents adjustment to the value of MtM account at the date t_l . In case when $\delta_{\text{MtM}}(t_l) < 0$ seller of the option withdraws the sum $\delta_{\text{MtM}}(t_l)$ from the MtM account while if $\delta_{\text{MtM}}(t_l) > 0$ the seller of the option should add the sum of $\delta_{\text{MtM}}(t_l)$ to MtM account. Now let us assume that seller of the call option defaults at a moment t_l , $t_l \leq T$, i.e.

$$S(t_l) - K \geq Q$$

Let us suppose that recovery value is a portion $q \in [0, 1)$ of the option value $S(t_l) - K > 0$. From option seller (B) perspective the cash flow to MtM account can be represented as following

$$\begin{aligned} \text{CF}_{\text{MtM}, B} = & \sum_{l=1}^L \chi(\tau = t_l) \left\{ \sum_{k=1}^{l-1} \delta_{\text{MtM}}(t_k) \chi(t = t_k) + \text{MtM}(t_l) - \right. \\ & - q[S(t_l) - K] \chi(t = t_l) \left. \right\} + \chi(\tau > T) \left\{ \sum_{k=1}^L \delta_{\text{MtM}}(t_k) \chi(t = t_k) + \right. \\ & + [\text{MtM}(T) - (S(t_l) - K)] \chi(t = T) \left. \right\} \end{aligned} \quad (13')$$

From option buyer (A) perspective the cash flow is different

$$\begin{aligned} \text{CF}_{\text{MtM}, A} = & -C(t_0, x) \chi(t = t_0) + \sum_{l=1}^L \chi(\tau = t_l) q[S(t_l) - K] \chi(t = t_l) + \\ & + \chi(\tau > T) [K - S(T)] \chi(t = T) \end{aligned} \quad (13'')$$

Uncollateralized cash flow of the option from the seller perspective with counterparty risk can be written in the form

$$\text{CF}_B = C(t, x) \chi(t = t_0) - \sum_{l=1}^L \chi(\tau = t_l) Q \chi(t = t_l) + \chi(\tau > T) [K - S(T)] \chi(t = T)$$

and uncollateralized cash flows from seller and buyer perspectives are equal in value, $CF_B = -CF_A$. Hence MtM adjustment is equal to

$$\begin{aligned} CF_{MtM, B} - CF_B &= \Sigma_0 - \Sigma_a \\ CF_{MtM, A} - CF_A &= -\Sigma_a \end{aligned} \tag{14}$$

where

$$\begin{aligned} \Sigma_0 &= \chi(\tau > T) MtM(T) \chi(t=T) - C(t, x) \chi(t=t_0) \\ \Sigma_a &= \sum_{l=1}^L \chi(\tau = t_l) \sum_{k=1}^{l-1} \delta_{MtM}(t_k) \chi(t=t_k) + \chi(\tau > T) \sum_{k=1}^L \delta_{MtM}(t_k) \chi(t=t_k) \end{aligned}$$

Formulas (14) represent MtM adjustment value in call option pricing for each market scenario. Assume for simplicity that buyer and seller of the option agree to use Black Scholes price $c_{BS}(t, S)$ to specify the call option price. Actually option buyer and seller can use different models of option price. In this case values of the MtM adjustments should be different for counterparties. Applying $c_{BS}(t, S)$ we will arrive at formulas (13'), (13''). One can use expected value of the present value of the cash flows to estimate the effect of the MtM account on spot pricing. Recall that a single number using reduction of the stochastic cash flow implies market risk, which specifies whether counterparty pays higher or lower price than it implies by realized market scenario. On the other hand MtM collateralization does not completely eliminate counterparty risk. If during a particular day underlying of the call option increases significantly the seller of the option may declare default. Nevertheless the use of the MtM account decreases losses of the option buyer in case of default or avoids default.

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